Aman Kumar

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EXPERIENCE

NativeLink **Remote (Palo Alto, US)**
Software Engineer *June 2025 – Present*

- Built and optimized **distributed systems in Rust**, including integrating Google Cloud Storage as a CAS backend to support large-scale caching and remote execution.
- Improved the **NativeLink scheduler** at the code level, significantly increasing its performance and horizontal scalability for large distributed build workloads.
- Upgraded and maintained **NativeLink's production-grade Helm chart** for Kubernetes (GKE), ensuring reliable, scalable, and automated deployments.
- Developed complete **Bazel remote execution** setups for enterprise clients, including build configuration, validation, and CI/CD integration to streamline onboarding and accelerate evaluation.

Fortanix Inc. **Bengaluru, India**
Software Developer II *Feb 2025 – June 2025*

- Designed a **distributed caching layer**, improving scan performance and significantly reducing service latency.
- Developed cryptography-focused tooling, including a **Post-Quantum Cryptography (PQC) Readiness Dashboard** and a **CBOM Export Tool** to analyze cryptographic usage and ensure compliance with secure key-management standards.
- Contributed to secure key-management and **enclave-based workflows** (Intel SGX, Azure, AWS), supporting production-grade deployments of the Data Security Manager.
- Increased test coverage from **32%** to **66%**, improving reliability through comprehensive unit and integration testing.

Software Developer I *Mar 2024 – Feb 2025*

- Optimized scan workflows by refining streaming patterns and enhancing the in-house **Azure SDK**.
- Implemented core features including **On-Premise Scan**, **Federated Authentication**, and **Policy Center APIs**.
- Expanded unit/integration test coverage for scan flows and CRUD APIs.
- Developed high-performance backend services in **Rust** and **Python** with a focus on secure, scalable cryptographic workflows and distributed systems.

Software Development Intern *Sep 2023 – Mar 2024*

- Extended the in-house **Azure SDK** with hierarchy and IAM support, enabling scalable cloud resource scanning.
- Contributed to CRUD API design and integration for Key-Insight.
- Implemented and secured cryptographic key management workflows in Data Security Manager, working with **Intel SGX enclaves**, **cloud HSMs** (Azure, AWS), and secure storage.

iRage Capitals **Mumbai, India**
Quantitative Analyst Intern *Dec 2022 – May 2023*

- Built a **real-time risk management system** with alert dashboards to monitor trading activity and threshold breaches.
- Analyzed derivatives data (PnL, IV Gap) to identify trading patterns and improve performance.
- Automated trading workflows using **Shell Scripting**, **Linux**, and **Python** (data pipelines, analytics, and report generation).

GNOME Foundation (*Google Summer of Code*) **Remote (Berlin, Germany)**
Software Development Intern *May 2022 – Sep 2022*

- Reduced application startup time by **32%** by optimizing database migration logic.
- Contributed automation scripts in **Python** to streamline testing and development workflows.
- Added **multi-user functionality** by migrating schemas to Tracker DB.
- Optimized performance of a Linux application stack using **Rust**, **GTK**, and **TrackerDB**.

PROJECTS

Order Matching Engine | *GitHub* **May 2023**

- Implemented a NASDAQ ITCH–compatible **Limit Order Book** in Rust, designed for ultra-low-latency, high-throughput electronic trading workloads.
- Achieved peak performance of **11.4M messages/sec** with end-to-end processing latency as low as **87 ns/tick**.
- Optimized memory layouts, lock-free data structures, and concurrency primitives to deliver consistent, predictable performance under extreme load.

Distributed Cache System | *GitHub* **Jan 2025**

- Built a fault-tolerant, scalable distributed cache in **Rust** using **consistent hashing** and **LRU/LFU** eviction.
- Implemented replication, leader/follower coordination, and resilient **failure-recovery** while exploring consensus trade-offs.
- Designed **thread-safe concurrent operations** with optimized memory layouts for predictable low-latency performance.

EDUCATION

Indian Institute of Information Technology, Nagpur **Nagpur, India**
Bachelor of Technology - Computer Science and Engineering *Jul 2019 - Jun 2023*

TECHNICAL SKILLS

Languages: Rust, C/C++, Python, SQL, JavaScript/TypeScript

Backend & Systems: Distributed Systems, Concurrency/Parallelism, Caching, Cryptography, REST/gRPC APIs, Tokio, Actix, Flask, Django, Cryptographic Systems, SGX, Confidential Computing

DevOps & Cloud: Bazel, Kubernetes (GKE), Helm, Docker, CI/CD, Google Cloud (GCS, GKE), Azure, AWS

Tools & Platforms: Linux, Git, Shell Scripting, GTK